

Toronto, ON
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Ruth Zhang

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EDUCATION

University of Toronto

Toronto, ON

Mathematical Applications in Economics & Finance Specialist, Statistics Major

Sept 2022 –

- **GPA: 3.99/4.0** (95/100)
- **Relevant Coursework:** Real Analysis, Differential Equations, Multivariable Calculus, Linear Algebra, Optimization, Probability, Statistics, Programming, Microeconomics, Macroeconomics, Econometrics, Finance

SKILLS

Programming Languages: Python, R, SQL, JavaScript

Technologies: MySQL, Git, GitHub, HTML/CSS, $\text{\LaTeX}$, Microsoft Office Suite (Excel, PowerPoint, Word), Airtable

EXPERIENCE

Research Assistant | University of Toronto

Toronto, ON

Supervisor: Avi Goldfarb (Rotman Chair in Artificial Intelligence & Healthcare)

Jan 2025 – Present

Illusion of Competition Project (Sep 2025 – Present)

- Designed R workflows (version-controlled via GitHub) to integrate large-scale consumer datasets, analyzing brand ownership structures and market dynamics across product categories.
- Consolidated and prepared survey and experimental data into a unified dataset of more than 100,000 observations, enabling empirical analysis of consumer misperception of competition.

Lab for the Management of Science and Technology (May 2025 – Aug 2025)

- Performed Exploratory Data Analysis (EDA) in R on over 5,000 procurement records from the Acceleration Consortium's expenditure dataset to identify spending patterns and support data-driven insights.
- Executed detailed visual classification of over 1,500 retail storefront images, generating a structured dataset that advanced empirical research on the relationship between storefront design and passerby foot traffic.
- Conducted preliminary literature review by identifying and summarizing 500+ entrepreneurship papers from leading academic journals that led to a published study on quantum computing as an enabling technology.

Creative Destruction Lab (Jan 2025 – Apr 2025)

- Labeled 500 (1% stratified random sample) open-text startup objectives to validate a zero-shot classification Machine Learning algorithm (GPT o3-mini), achieving 71% agreement and confirming model reliability.
- Cross-referenced over 300 funding round records among early-stage, science-based startups at CDL to enable accurate linkage to an external data source, enhancing the overall dataset for analysis.

Teaching Assistant | University of Toronto

Toronto, ON

Courses: MAT135 – Calculus I, MAT136 – Calculus II

Sept 2024 – Apr 2025

- Conducted weekly tutorials for classes of 30 students, clarifying complex mathematical concepts.
- Collaborated with course instructors to design and implement strategies for improving student participation.

E-commerce Partner

Remote

Freelance

Jan 2020 – Present

- Built and scaled a digital business by designing and marketing handwritten fonts, achieving 10,000+ sales.
- Collaborated with leading stationery brands (e.g., Zebra) on high-visibility co-branded campaigns.